



An automatic and non-invasive physical fatigue assessment method for construction workers

Yantao Yu^a, Heng Li^a, Xincong Yang^{a,*}, Liulin Kong^b, Xiaochun Luo^a, Arnold Y.L. Wong^c

^a Department of Building and Real Estate, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong SAR, China

^b Department of Civil Engineering and Mechanics, Huazhong University of Science and Technology, Wuhan, China

^c Department of Rehabilitation Sciences, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong SAR, China

ARTICLE INFO

Keywords:

Occupational safety and health
Construction worker
Ergonomic
Deep learning
Machine learning
Computer vision

ABSTRACT

The construction industry around the globe has unsatisfactory occupational health and safety records. One of the major reasons is attributed to high physical demands and hostile working environments. Construction work always requires workers to work for a long duration without sufficient breaks to recover from overexertion and to work under harsh climatic conditions and/or in confined workspaces. Such circumstances can increase the risk of physical fatigue. Traditionally, fatigue monitoring in the construction domain relies on self-reporting or subjective questionnaires. These methods require the manual collection of responses and are impractical for continuous fatigue monitoring. Some researchers have used on-body sensors for fatigue monitoring (such as heart rate monitors and surface electromyography (sEMG) sensors). Although these devices appear to be promising, they are intrusive, requiring sensors to be attached to the worker's body. Such on-body sensors are uncomfortable to wear and could easily cause irritation. Considering the limitations of these methodologies, the current research proposes a novel non-intrusive method to monitor the whole-body physical fatigue with computer vision for construction workers. A computer vision-based 3D motion capture algorithm was developed to model the motion of various body parts using an RGB camera. A fatigue assessment model was developed using the 3D model data from the developed motion capture algorithm and biomechanical analysis. The experiment showed that the proposed physical fatigue assessment method could provide joint-level physical fatigue assessments automatically. Then, a series of experiments demonstrated the potential of the method in assessing the physical fatigue level of different construction task conditions such as site layout and the work-rest schedules.

1. Introduction

Construction industry around the globe is distressed with unsatisfactory occupational health and safety records. One of the major reasons behind the health and safety issues is the highly physical and demanding nature of the construction tasks. Construction related work has to be done for prolonged duration of work-time without sufficient break to recover from overexertion, in harsh climatic conditions and in confined work-spaces. Such circumstances may expose construction workers to the risk of fatigue development. If workers continue to work under fatigued condition, they are prone to development of work-related musculoskeletal disorders (MSDs), make more mistakes during work, degradation of the workmanship, reduction in productivity, accidents on the construction sites and fall incidents. According to Bureau of Labor Statistics [1], 33% of all occupational injuries and illnesses on

the US construction sites were related to fatigue and overexertion. Besides, hot and humid conditions may further accelerate the fatigue development which may cause the construction workers to suffer from heat stroke and might even cause death. During 2010, heat strokes arising from hot and humid environments claimed 47 deaths in the Japanese industrial sector [2].

The construction industries, especially those in developed countries or regions, should pay closer attention to fatigue detection and prevention because of several labor issues which may hinder the development of the construction sector. These issues include rising labor wages, ageing workforce and a shortage of manpower. In 2013, the proportion of the construction workforce in Hong Kong over the age of 50 years was 44% whereas that of 60 years was 12% [3]. Older workers are more prone to fatigue because of decline in physical work capacity and muscle strength. Moreover, there is a consistent expected shortfall

* Corresponding author.

E-mail addresses: yt.yu@connect.polyu.hk (Y. Yu), bshengli@polyu.edu.hk (H. Li), 14902497r@connect.polyu.hk (X. Yang), kellykong@hust.edu.cn (L. Kong), bsericlo@polyu.edu.hk (X. Luo), arnold.wong@polyu.edu.hk (A.Y.L. Wong).

<https://doi.org/10.1016/j.autcon.2019.02.020>

Received 21 August 2018; Received in revised form 17 February 2019; Accepted 22 February 2019

0926-5805/ © 2019 Published by Elsevier B.V.

of 10,000 to 15,000 construction workers from 2017 to 2021 [4]. Under such circumstances, it becomes necessary to make the current construction workforce sustainable by improving their occupational health and safety. Since, fatigue poses a major challenge to occupational health and safety, fatigue monitoring and management has become an issue of utmost prominence.

Traditionally, fatigue monitoring in the construction domain has relied on subjective information collected using questionnaires [5,6]. As such, these methods depend on manual collection of responses and are not practical for continuous fatigue monitoring. Other researchers have tried to make use of on-body sensors for fatigue monitoring [2,7–10]. Although these studies have proven the concept, all of them are intrusive in nature, requiring multiple sensors to be attached to the body of construction workers. Such on-body sensors are not comfortable to wear and could easily instigate irritation. Considering the limitations of the aforementioned methodologies, this research proposed a blend of computer vision technology and biomechanical analysis for fatigue monitoring of the construction workers. The proposed method could provide a novel non-intrusive method to monitor whole-body fatigue. First, a computer vision-based 3D motion capture algorithm was developed that could model the motion of various body parts while performing construction tasks using an RGB camera. Together with the exerted forces data, inverse dynamics was applied to calculate the joint-level torque. Finally, a fatigue assessment model was applied to enable the individualized fatigue monitoring.

For the first time, it is realized to automatically assess construction workers' physical fatigue on joint-level according to the data from real construction sites. This technology will reduce the possibility of over-exertion and fatigue among workers, and overall will improve occupational health and safety of the construction industry. Development of such a system will equip the industry with a non-invasive tool for fatigue monitoring, which could also be used in the other industries.

2. Literature review

Fatigue means reduced functional capacity and tiredness that occurs during and at the end of the workday [11]. Fatigue involves physical fatigue, mental fatigue and emotional fatigue. This study focuses on physical fatigue. Numerous methods have been proposed to measure physical fatigue level, including subjective methods and objective methods. Subjective methods rely on workers' self-report of their physical fatigue perception. Borg RPE scale and Borg CR10 scale, for example, ask the workers to report their perception of fatigue as a score from 6 to 20 or from 0.0 to 12.0 [12]. Need for Recovery Scale (NRS) asks 11 questions to assess the workers' demand for rest [13]. Fatigue Assessment Scale is a 10-item scale including 5 questions about physical fatigue [14]. Self-report methods are easy to implement but not suitable to investigate construction worker's physical fatigue. The reasons include 1) The report process may interrupt normal work activity; 2) Self-report only provide the final fatigue status, so the data cannot support the investigation of physical fatigue development; 3) The report data may be inaccurate due to the ambiguous score standard or workers dishonesty. Based on the above consideration, this study aims to develop an objective physical fatigue assessment method that is non-intrusive, continuous and accurate. The rest of this section is a review on previous objective physical fatigue assessment methods.

2.1. Physiological indicators

Physical fatigue is essentially a physiological process, which can thus be assessed with physiological indicators including cardiovascular indicators and electronic indicators. Cardiovascular indicators include heart rate, skin temperature and breathing rate, which usually increase under high physical strain [10,15,16]. The indicators are measured with sensors, which are usually tied tightly to worker's body to ensure the accuracy and may hinder the workers during work routine. In addition,

the sensors need to be charged every “several hours” within a day, making it difficult to monitor physical fatigue in longer periods.

A widely-used electronic indicator is surface electromyographic (sEMG). sEMG measures the electric activity of groups of muscles with electrodes placed on the skin surface of the target muscle groups. sEMG has been used in the construction industry to measure worker's muscle fatigue [7,17]. Though sEMG can accurately measure muscle fatigue, the method cannot be applied on the construction site, because the sEMG electrodes need to be attached directly to the skin. In addition, it requires at least two electrodes to measure the level of a muscle group. As a result, a lot of electrodes need to be attached to worker's skin to measure the whole-body fatigue, leading to high cost and hindering worker's normal work.

In summary, physiological indicators can assess physical fatigue accurately. However, the physiological indicators require the workers to wear certain sensors, which is not suitable for long-period physical fatigue assessment on construction site, because 1) the sensors may be uncomfortable to wear; 2) the sensors may require being charged frequently.

2.2. Biomechanical methods

Biomechanical methods measure physical fatigue from the perspectives of external loads and working postures. These factors are directly related to construction site management such as site layout and work-rest schedule. Numerous studies have applied biomechanical methods to assess workers' physical fatigue. This section reviewed the previous studies on biomechanical analysis methods and data collection methods.

2.2.1. Biomechanical analysis method

2.2.1.1. Posture-based methods. Working posture is a critical biomechanical factor leading to physical fatigue. Previous studies assessed construction workers' physical fatigue risks using posture data. Some studies assessed physical fatigue with joint angles. Given the joint angles (e.g. trunk flexion/bend angle, shoulder flexion/abduction angle and elbow/knee flexion angle), these methods divided the angles into different ranges and gave physical fatigue scores based on the ranges [18–20]. Other studies first identified working postures (e.g. standing up, back bending, squatting, etc.), then provided physical fatigue assessment based on the postures [21]. A series of field experiments validated the feasibility of the methods. However, in addition to awkward postures, lifting weights (materials or heavy tools) is also a common-seen factor of physical fatigue. Above methods only focused on posture data, which may lead to the underestimation of physical fatigue.

2.2.1.2. Posture and exerted force-based methods. Some ergonomic scales assess physical fatigue considering both postures and exerted forces, including Key Item Method (KIM), NIOSH lifting equation, Ovako Working Posture Analysis System (OWAS), OCRA (Occupational Repetitive Actions), RULA (Rapid Upper Limb Assessment) and REBA (Rapid Entire Body Assessment) [22–27]. When assessing posture factors, these scales use the similar ways with the posture-based methods. When assessing exerted forces, these scales divided the forces into different categories according to the absolute value, such as 0–5 kg, 5–10 kg and over 10 kg, and then provide an exerted force score based on the category. Finally, a physical fatigue score is provided by combining the posture-based score and the exerted force-based score. In addition, some scales also consider work pattern. For example, RULA and REBA consider the work repetitiveness and duration. The final physical fatigue score will be higher if a task is repeated more than four times per minute or lasts for more than a minute.

These tools are easy to use and provide quantitative workload assessment but may face with the following challenges if applied in the construction industry: 1) some tools only focus on certain body parts,

e.g. OCRA focuses on the upper body only; 2) these tools are aimed at manufacturing workers and consider motion and external load as repetitiveness, while the work of construction workers is less regular, making it difficult to assess physical fatigue based on repetitiveness or duration. 3) Assessing the same workload with different scales may lead to different results [28]. To solve these issues, this study aims to develop a physical fatigue assessment method that is suitable for the complex nature of construction activities by using biomechanical calculations. The following section reviews the biomechanical calculation-based methods available.

2.2.1.3. Biomechanical calculation-based methods. Given exerted forces and postures, biomechanical analysis can measure physical fatigue in terms of joint force or torque. Several physical fatigue assessment tools have been developed based on biomechanical calculations, including 3DSSPP, OpenSim and AnyBody Modelling System [29–31]. More specifically, 3DSSPP estimates the joint-level workload, while OpenSim and AnyBody focus on muscle-level workload. In addition, 3DSSPP only works in static cases, while the other two theoretically work in any case. 3DSSPP and OpenSim have been applied in the construction industry to assess workers' ergonomic risks, demonstrating the feasibility of applying biomechanical calculation to assess workers' physical fatigue [32,33]. However, as analyzed in the following part, biomechanical calculation methods can only be used in virtual or indoor environments, limiting the application on construction sites [32,33].

2.2.2. Data collection methods

Manual observation has been widely used to collect motion data in previous studies [34,35]. However, manual observation may have the following challenges if applied on construction fields. 1) The observation results depend on the observer's experience and subjective standards, thus may be not accurate enough to support biomechanical calculation; 2) it is time- and labor-consuming to collect data and assess physical fatigue level through human observation [36].

Inertial measurement unit (IMU) has been widely used to obtain construction worker's motion data for occupational health analysis. IMU sensors can obtain joint angle data automatically and accurately. Based on the joint angle data and the posture-based methods, the physical fatigue level can be assessed automatically [10,20,21]. The main disadvantage, however, is the requirements to tie IMU sensors tightly to the human body – an intrusiveness that may not be acceptable in practice. In addition, such sensors may be feasible for a short-period but are likely to be irritating in long-time application [36,37].

Motion capture systems, such as VICON system, provide highly accurate 3D motion data. The systems require that markers be placed on the subjects' body and cameras be placed in the surroundings. Through processing the received signals from the markers, the system can calculate the 3D position of each marker, and thus facilitate behavior-based management [38]. The accuracy of such systems is as high as millimeter [39]. Such systems are usually used together with biomechanical calculation software, such as OpenSim, to provide muscle-level fatigue analysis [30]. However, the system only works in laboratory environments, because the system requires at least 4 cameras within 10 m and more than 10 markers to capture a worker's whole-body posture, which is thus impractical to be applied on construction site.

Vision-based motion capture methods provide a non-intrusive solution for working motion data collection. Depth cameras have been proved to be an effective tool to collect construction workers' motion data in indoor environments [33,40,41]. The accuracy of depth cameras, however, might be very low in outdoor environments due to the interference of the sunlight to the infra-red signal. To solve this problem, researchers have tried to extract working postures from RGB images captured by ordinary cameras. With the help of machine learning algorithms, construction workers' 2D postures have been

successfully extracted from RGB images [19,42]. The output of 2D posture recognition methods is posture classification (squatting, bending or lifting) or joint angle ranges instead of absolute joint angles. As a result, the posture recognition result can only be used in posture-based ergonomic analysis, rather than detailed physical fatigue assessments.

To realize detailed physical fatigue assessment on construction sites, it is important to develop an accurate 3D motion capture method that is non-intrusive and suitable for outdoor environment. The single-lens 3D posture estimation algorithm can be used to realize the aim [43]. In this study, we first examined the accuracy of the single-lens 3D posture estimation algorithm when applied to construction tasks. Then, the 3D posture data, together with exerted force data, was fed to biomechanical calculations.

2.3. Muscle development models

In biomechanics, several muscle fatigue models have been built to predict muscle-level fatigue according to the fatigue development mechanism. For instance, a muscle fatigue level was proposed based on Ca^{2+} cross-bridge mechanism, which focusses on the muscle fatigue development process of quadriceps [44]. Though the model can predict muscle fatigue correctly, it is not suitable for industrial application due to the high complexity (more than 20 variables just for quadriceps). Another muscle fatigue model was developed based on the muscle force-pH relations [45]. However, this model only predicts the maximum muscle force capacity, instead of analyzing muscle force development during working. A new model was proposed to predict physical fatigue based on joint torque [46]. The model was first theoretically built based on the muscle motor fatigue development theory, then validated with a series of experiments [46,47]. The model was successfully applied in virtual construction environments to assess physical fatigue [32]. The gap between the model and field application is the lack of a suitable 3D motion capture method and the following biomechanical calculations. This research aims to fill the gap by merging the state-of-art motion-capture algorithms and biomechanical analysis.

2.4. Summary

Table 1 compares all the above physical assessment methods and data collection methods from the aspects of assessment content, accuracy, focused body parts, work pattern, suitable environments and invasiveness. The aim of this study is to provide an accurate and non-invasive physical fatigue assessment method suitable for construction field environments. Based on the comparison result of the table, the combination of the muscle motor unit model and the 3D posture estimation algorithm could reach the goal. The methodology is detailed illustrated in Section 3, and then validated by the experiments in Section 4.

3. Methodology

The research aims to develop a new quantitative physical fatigue evaluation method for construction workers. Fig. 1 depicts the flow chart of the proposed construction worker's fatigue evaluation method, which contains three steps. In data collection period, the method collects 3D motion data automatically and non-invasively. Then given the exerted forced data and human body parameters, the joint torques are calculated through biomechanical calculations. Finally, a physical fatigue index is given based on the muscle fatigue development model.

3.1. 3D motion capturer from 2D video

To make the proposed method non-intrusive, and to provide 3D motion data for the biomechanical calculation period, the 3D working motion data must be collected automatically without interfering work

Table 1
The comparison of physical fatigue assessment methods and data collection methods.

Assessment method		Content	Data collection methods		Accuracy	Body parts	Work pattern	Environment	Invasiveness
Subjective	Self-report	Self-perception of physical fatigue	Questionnaire interview	Low (subjective data)	✓	✓	✓	✓	Intermittent recording procedures
Objective	Physiological indicators	Physical reflections due to physical fatigue	PSM Sensor	✓	✓	✓	✓	✓	Uncomfortableness due to the sensor
		Muscle electrical reflections due to physical fatigue	sEMG Sensors	✓	✓	Low back	✓	Lab	Uncomfortableness due to the electrodes
	Mechanical indicators	Ergonomic risks due to awkward postures, large exerted forces, and long or repetitive task patterns	Observation	Low (observer variance)	✓	✓	Repetitive and simple	✓	✓
			Wearable sensor	✓	✓	✓	✓	✓	Uncomfortableness due to the sensor
			Depth Camera	✓	✓	✓	✓	Indoor	✓
			RGB camera + 2D posture data	Low (Only posture classification)	✓	✓	✓	✓	✓
			Motion capture system	✓	✓	✓	✓	Lab	More than ten markers and four cameras
	Biomechanical analysis	Muscle- or joint-level physical fatigue	-	✓	✓	Quadriceps	✓	Lab	-
	Ca ²⁺ cross-bridge model	Muscle-level physical fatigue	-	✓	✓	-	✓	Lab	-
	Muscle force-pH model	Maximum muscle capacity	-	✓	✓	-	✓	Lab	-
	Muscle motor unit model	Muscle- or joint-level physical fatigue	RGB camera + 3D posture data	✓	✓	✓	✓	✓	✓

activities, while ensuring data accuracy. A single RGB camera-based 3D motion capture algorithm is involved to solve this problem [43]. Fig. 2 is the framework of the algorithm.

The algorithm first identifies key joints (neck, shoulders, elbows, wrists, hips, knees and ankles) and the locations of the joint-related pixels in the 2D RGB images using Stacked Hourglass Networks [48]. The network is composed by multiple hourglass modules placed end to end. Each hourglass module is a convolutional neural network, which extracted features from a RGB image through convolution operations. An hourglass module first pools down the original RGB image to different resolutions, then extracts and combines features across multiple resolutions, so that the module considers both local and global features, corresponding to the joint information and the whole-body information in our case.

Given the 2D locations of each joints, another network is trained to infer the 3D locations of each joint [43]. The basic assumption of the network is that the ratio of two bones of the same person should be a constant. The network is trained on MPII dataset, which contains the 2D and 3D pose data of postures such as squatting, standing and walking [49]. According to the assumption and the subject's height data, the network is trained to minimize the differences of bone length ratios between the inferred 3D postures and the ground truth 3D postures in the MPII dataset.

3.2. Inverse dynamics for joint-level torque analysis

The biomechanical structure of human is complex, since the mechanical properties of bones, joints, tendons and muscles are varied with the individual factors such as age, gender, weight and height. In addition, the stress-strain relations of bones, joints, tendons and muscles are also varied with the exerted forces. To facilitate the internal load analysis, this research makes the following assumptions: 1) the human body is simplified as a lever system connected with hinge joints; 2) each lever is a rigid, which means the length is a constant regardless of the external load; and 3) workers' motions on construction sites are slow and steady, which means the lever system is in an equilibrium status. Based on the above assumptions, the force balance equation and torque balance equation are used to solve the joint torques. Fig. 3 presents the simplified human skeleton model for biomechanical analysis, which contains 15 key joints including the torso and four limbs. The coordinates are a right-hand system. The positive direction of the y axis is straight up. Fig. 4 shows the force and torque of a certain segment.

3.2.1. Computing joint reaction forces

Newton's equation is used to calculate joint reaction forces. For a certain segment, the equation can be expressed as Eq. (1).

$$F_{Pr} + F_{Ch} + G = 0 \quad (1)$$

where,

$F_{Pr} = (F_{Pr, x}, F_{Pr, y}, F_{Pr, z}) \in \mathbb{R}^3$ is the joint reaction force at the parent joint [N];

$F_{Ch} = (F_{Ch, x}, F_{Ch, y}, F_{Ch, z}) \in \mathbb{R}^3$ is the joint reaction force at the child joint [N];

$G = (0, -mg, 0) \in \mathbb{R}^3$ is the gravity of the segment [N]; m is the mass of the segment [kg]; $g = 9.8 \text{ m/s}^2$;

The positive direction is straight up, which is the same with the positive direction of y axis in Fig. 3.

In this research, the mass of each segment, m is calculated based on the segment percentage of total body weight, which can be referred to [50]. Specifically, according to the skeletal model in Fig. 3, there is no child joint reaction force in the force equilibrium equations of lower arms and shanks. The forces are replaced with ground reaction force and hand load forces due to the tools or materials hold in hands. In this research, the hand load force is assumed to be the weight of the tools and materials, and the ground reactions force is assumed to be the sum

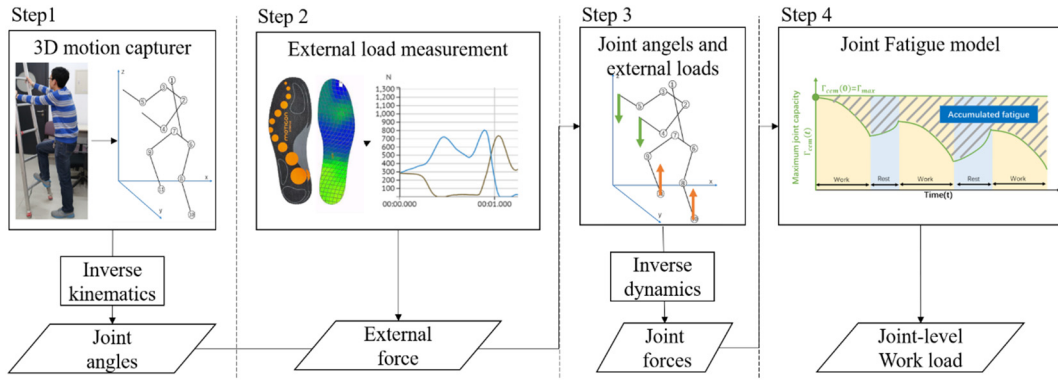


Fig. 1. An overview of the physical fatigue assessment method.

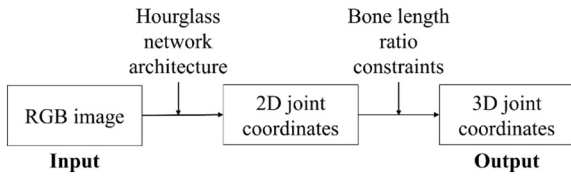


Fig. 2. The single RGB camera-based 3D motion capture algorithm.

of hand load force and the subject's weight. Given the ground reaction forces, the joint reaction forces of knees can first be figured out, then the joint reaction forces of other joints can be figured out hieratically.

3.2.2. Computing joint torques

Torque balance equation is used to calculate joint torques. Fig. 4 shows the torques on a certain segment, where A is the parent joint, B is the child joint, and C is the gravity center point. The sum of all the torques acting on the parent joint equals zero. The torques include parent joint torque, child joint torque, and the torques generated from the segment's self-weight and the joint reaction force on the child joint. The torque generated from the parent joint reaction force is zero. Eq. (2) is the joint moment balance equation.

$$T_{pr} + T_G + T_{F_{Ch}} + T_{Ch} = 0$$

$$T_G = \overrightarrow{AC} \times \mathbf{G} = r \overrightarrow{AB} \times \mathbf{G}$$

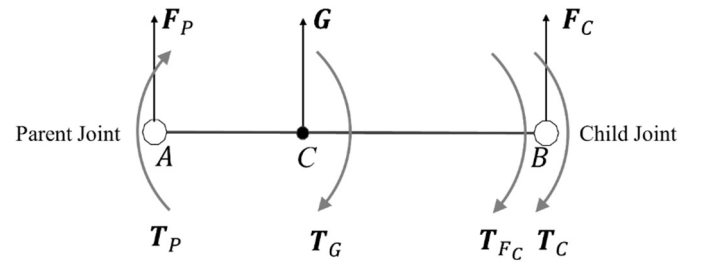


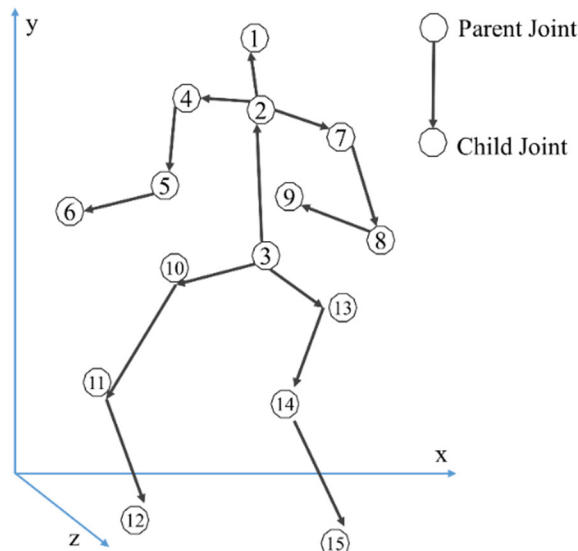
Fig. 4. Forces and moments of a given segment.

$$T_{F_{Ch}} = \overrightarrow{AB} \times F_{Ch} \quad (2)$$

where,

T_{pr} is the reaction torque at the parent joint A; T_G is the torque produced by $\mathbf{G}$; $T_{F_{Ch}}$ is the torque produced by F_{Ch} ; T_{Ch} is the reaction force at the child joint B. The unit of torque is [N·m]. The positive direction is clockwise.

$\overrightarrow{AC}$ is the vector from the parent joint to the center of mass of the segment; $\overrightarrow{AB}$ is the vector from the parent joint to the child joint; r is the ratio of $\overrightarrow{AC}$ to $\overrightarrow{AB}$, which represents the location of the center of mass. The value of r is given in [50].



No.	Joint
1	Head
2	Neck
3	Waist
4	Right Shoulder (RS)
5	Right Elbow (RE)
6	Right Wrist (RW)
7	Left Shoulder (LS)
8	Left Elbow (LE)
9	Left Wrist (LW)
10	Right Hip (RH)
11	Right Knee (RK)
12	Right Ankle (RA)
13	Left Hip (LH)
14	Left Knee (LK)
15	Left Ankle (LA)

Fig. 3. The simplified biomechanical human skeleton model.

3.3. Joint level physical fatigue assessment

This module aims to identify physical fatigue according to the current loads on joints, and the load history of these joints. The fatigue and recovery model developed by Ma et al. [46] is applied to predict construction workers fatigue together with the posture and pressure data.

Fatigue is basically depended on 1) the intensity, duration and repetitiveness of external work load and 2) the personal muscle strength capacity of the individual. The fatigue development speed is positively correlated to the external load, and negatively correlated to muscle strength capacity. The model expressed the above correlations as Eq. (3). $F(t)$ represents fatigue level. Γ_{max} represents the maximum joint capacity. $\Gamma_{cem}(t)$ represents the current joint capacity, which decreases during working and increase during rest. $\Gamma(t)$ means the joint torque. In Eq. (3), $\frac{\Gamma_{max}}{\Gamma_{cem}(t)}$ is the reciprocal of current relative joint capacity, representing the personal factors. $\frac{\Gamma(t)}{\Gamma_{cem}(t)}$ is the current relative joint load, representing the external factors.

$$\frac{dF(t)}{dt} = \frac{\Gamma_{max}}{\Gamma_{cem}(t)} \frac{\Gamma(t)}{\Gamma_{cem}(t)} \quad (3)$$

In Eq. (3), the Γ_{max} of people of different ages, gender and ethnicities is derived from Shaunak, Ang et al. (1987) and Meldrum, Cahalane et al. (2007), [51,52]. $\Gamma(t)$ is the result of biomechanical analysis. $\Gamma_{cem}(t)$, the current torque capacity will decrease under external load, which is thus decided by the torque work load history. Muscle motor unit theory is applied to simulate $\Gamma_{cem}(t)$ based on the joint load history and the joint capacity. According to the muscle motor unit theory, the muscles generate torque because of the activation of muscle motor units. Some units have a high muscle force generation capacity that decreases rapidly (easy fatigue). The others have a lower muscle force generation capacity that decreases slowly (fatigue resistant). When the muscles are under workload, the former type of motor units will be first activated, followed by the latter one. Based on this above theory, torque generation capacity decreases more rapidly under a higher workload. In addition, under the same workload, the decreased speed will become slower due to the working order of the two kinds of muscle motor units [53]. Eq. (4) depicts the above process, where k is a constant value, and equals to 1 min^{-1} .

$$\frac{d\Gamma_{cem}(t)}{dt} = -k \frac{\Gamma_{cem}(t)}{\Gamma_{max}} \Gamma(t) \quad (4)$$

$\Gamma_{cem}(t)$ can be calculated as the integral of Eq. (4), as shown in Eq. (5).

$$\Gamma_{cem}(t) = \Gamma_{cem}(t_0) \exp\left(-\frac{k}{\Gamma_{max}} \int_{t_0}^t \Gamma(u) du\right) \quad (5)$$

When certain body parts are in a state of rest, the capacity of related joints will recover. As is shown in Fig. 5, the joint capacity will increase when the workers are taking a rest. According to the muscle motor unit theory, the recover process is essentially the recovery of fatigue motor unit, which can be represented by $\Gamma_{max} - \Gamma_{cem}(t)$. The recover process can be expressed as Eq. (6), where R is set as 2.4 min^{-1} , which is the average rate of recovery according to Liu, Brown et al. (2002), [54].

$$\frac{d\Gamma_{cem}(t)}{dt} = R(\Gamma_{max} - \Gamma_{cem}(t)) \quad (6)$$

By integrating Eq. (6), the recovery of joint capacity is presented in Eq. (7).

$$\Gamma_{cem}(t) = \Gamma_{cem}(t_0) + (\Gamma_{max} - \Gamma_{cem}(t_0))(1 - e^{-Rt}) \quad (7)$$

Given $\Gamma_{cem}(t)$, the fatigue index $F(t)$ can be calculated as the integral of Eq. (1).

Finally, given $\Gamma_{cem}(t)$, the fatigue index $F(t)$ can be calculated as the integration of $dF(t)/dt$. Fig. 6 is an example of the fatigue index $FT(t)$,

which increases rapidly under work status due to the decrease of joint capacity and increases slowly or even decreases due to the increase of the joint capacity.

4. Experiments and results

A series of studies have proved the usefulness and validity of the 3D motion capture method [43] and the physical fatigue assessment model [47,55]. However, given the diversity of construction trades and the complex nature of construction fields, it is necessary to validate the methods with construction tasks. Two series of experiments are conducted to validate the accuracy of the 3D motion capture method and the usefulness of the physical fatigue assessment method.

4.1. Testing the accuracy 3D motion estimation method

A laboratory experiment was conducted to test the accuracy of the 3D motion estimation method by comparing the estimated 3D motion data from RGB images and the measured 3D motion data with IMU sensors. The subject simulated three construction activities including material handling, plastering and rebar. The subject was required to wear an IMU system (3-Space™ Wireless 2.4 GHz DSSS, OH, USA) to provide the ground truth for the 3D motion data. The IMU sensor has an accuracy of 1° [56]. The IMU system includes 15 sensors. Fig. 7 shows the experiment scenes. The experiment gathered 3878 frames of RGB images and 4654 frames of IMU data. The data from the two sources were first synchronized and then compared to assess the RGB image-based 3D motion recognition results. Table 2 is the error of the 3D motion capture method. The mean error of the 3D location of each joint is 3.90 cm, and the standard deviation is about 1.59 cm.

4.2. Testing the usefulness of the physical fatigue assessment method

This experiment aims to validate the usefulness of the physical fatigue assessment method through a scaffolding task and a masonry task.

4.2.1. Experiment design

In the scaffolding task, the subject was asked to build up a cube with two-meter-long steel tubes and couplers, as shown in Fig. 8. The construction site layout was shown in Fig. 9. The work area was the location of the cube. The storage area was the place to store the steel tubes and couplers. The straight-line distance from the storage area to the work area was about 6 m. During the experiment, the subject first carried a tube from the storage area to the working area, then installed the tube with couplers. Above process was repeated for 16 times to finish the task. Two RGB cameras (smartphone cameras) were fixed on tripods to record the subjects' motion, as shown in Fig. 9.

In the masonry task, the subject was asked to build a concrete block walls as shown in Fig. 10. Each cement block wall measured approximately xx height xx weight xx depth. Each concrete block measured xx height xx weight xx depth and weighted approximately 16 kg. The concrete blocks were placed 1 m away from the cement block wall, as shown in Fig. 11. During the experiment, the subject first bent to pick up a block, then turned around to lay the block. The motion was repeated until a course has been installed. Then the subject checked the course with a level and evened it out with a thicker layer of mortar. One RGB camera (smart phone cameras) was fixed on tripods to record the subjects' motion, as shown in Fig. 11.

4.2.2. Experiment data

4.2.2.1. *Original data.* The input data includes anthropological parameters (Table 3) and the video records of the two tasks (Table 4).

4.2.2.2. *Data process.* The anthropological parameters were used to estimate the maximum joint capacity, the body segment mass, and the location of center of mass of each body segment. To eliminate the

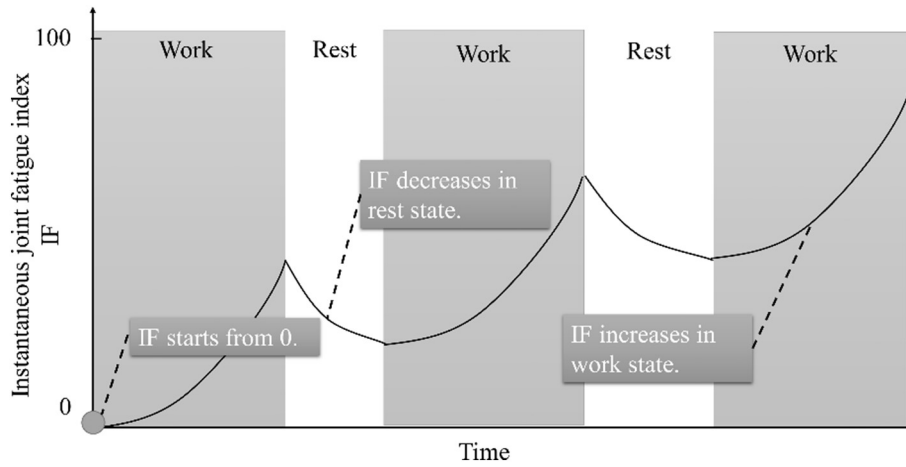


Fig. 5. An example of the instantaneous joint physical fatigue index in work and rest status.

effects of visual obstruction, two cameras recorded the subjects' motions simultaneously. The videos from two cameras were compared frame by frame and the one with fewer obstructions was selected to eliminate the effects of obstructions such as the scaffold and the concrete block wall.

4.2.3. Experiment results

Figs. 12a and 12b show the fatigue development of the scaffolding task, while Figs. 12c and 12d show the fatigue development of the masonry task. Both joint capacity and fatigue index were used to assess the fatigue level. The difference lies in that the joint capacity reflects the specific fatigue level of each frame (Eq. (5)), while the fatigue index reflects the accumulated fatigue level from the start to a certain frame (Eq. (7)).

Figs. 12a and 12c show the joint capacity of key joints during the scaffolding task and the masonry task respectively. The horizontal axis represents the frames and the vertical axis represents the joint capacity as the ratio of the current joint capacity to the maximum of the joint capacity. The ratio, instead of the absolute value of joint capacity, is used to eliminate the differences of the maximum joint capacity for the comparison among the joints. The eight curves correspond to eight key joints including left/right shoulders/elbows/hips/knees, as is shown in the figure keys. There was a generally decrease trend in the joint capacity during both tasks. It should be noted that there existed significant fluctuations in the joint capacity curves of hips and knees during the scaffolding task (Fig. 12a). The reason is that the joint capacity increased when the subject was in rest status. In the scaffolding

task, the subject needed to go back to the storage area with no burden after fixing a steel tube, which means that the subject's body segments were in a relax status with no awkward postures or external loads, so the joint capacity recovered.

Figs. 12b and 12d show the fatigue index of the key joint during the two tasks. The horizontal axis represents the frames and the vertical axis represents the fatigue index of each joint. All of the fatigue index curves during the two tasks show a continuous increase trend. The variance of the increasement reflects the variance of the fatigue development speed. When the joint capacity decreased quickly, the fatigue index increased sharply. On the contrary, when the joint capacity decreases slowly, the fatigue index increased slowly or even decreased.

As is shown in Figs. 12a–12d, the proposed physical fatigue assessment method can provide joint-level fatigue development analysis results. As Figs. 12a and 12b reveal, the subject's lower body (including both hips joints and both knees joints) had higher fatigue level than the upper body. Especially, the left hip and the right knee have the highest and the second highest fatigue level. This may suggest that the subject should pay more attention to the health of the two joints. In the masonry case (Figs. 12b and 12d), the subject's left knee and left hip had the highest and the second highest fatigue level. The final fatigue index of the left leg was about five times of the right leg. It may suggest that the subject in the masonry task should pay more attention to balance the workload of his legs.

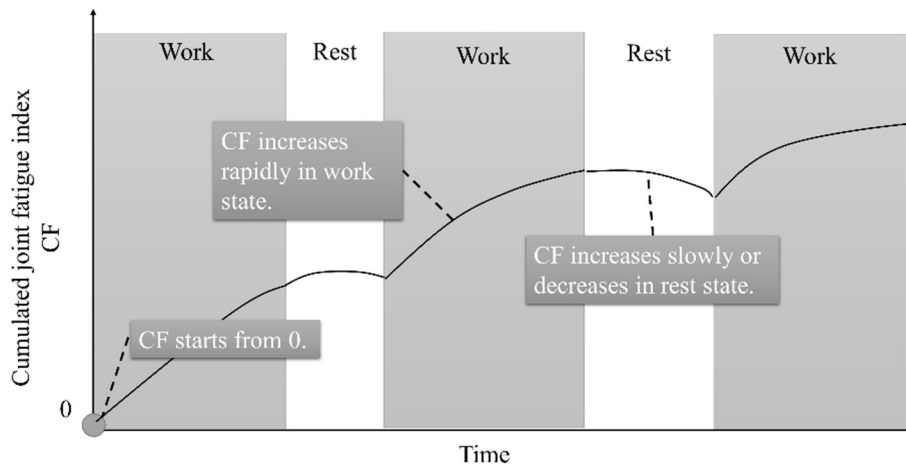


Fig. 6. An example of the cumulative joint physical fatigue index at the work and rest states.



Fig. 7. The inertial measurement unit sensors and the tasks of the laboratory experiment.

Table 2

The error of the 3D motion capture method unit: cm.

Task		Head	LS	RS	LE	RS	LW	RW	LH	RH	LK	RK	LA	RA	Mean
Rebar	Mean	6.12	4.17	3.11	4.76	4.38	5.40	5.79	0.88	0.88	0.88	1.86	3.62	4.10	3.53
	Std. ^a	1.99	1.22	1.19	1.15	1.11	1.24	1.10	0.32	0.32	0.32	0.89	2.21	2.45	1.40
Bricklayer	Mean	7.94	5.69	5.58	4.23	3.71	6.38	5.57	1.13	1.13	1.13	3.02	9.10	9.47	4.93
	Std.	1.94	1.36	1.25	0.88	0.86	1.02	1.27	0.35	0.35	0.35	1.20	1.62	1.66	0.93
Plaster	Mean	10.38	8.85	8.77	7.58	8.98	10.31	12.97	0.58	0.58	0.58	2.11	2.74	3.14	3.24
	Std.	1.45	1.01	1.31	0.93	1.95	1.07	2.00	0.17	0.17	0.17	1.00	1.22	1.24	1.32
Mean	Mean	8.14	6.24	5.82	5.52	5.69	7.36	8.11	0.87	0.87	0.87	2.33	5.16	5.57	3.90
	Std.	2.26	1.70	1.93	1.30	1.70	1.71	2.27	0.34	0.34	0.34	1.03	2.84	2.96	1.59

^a Std. represents standard deviation.

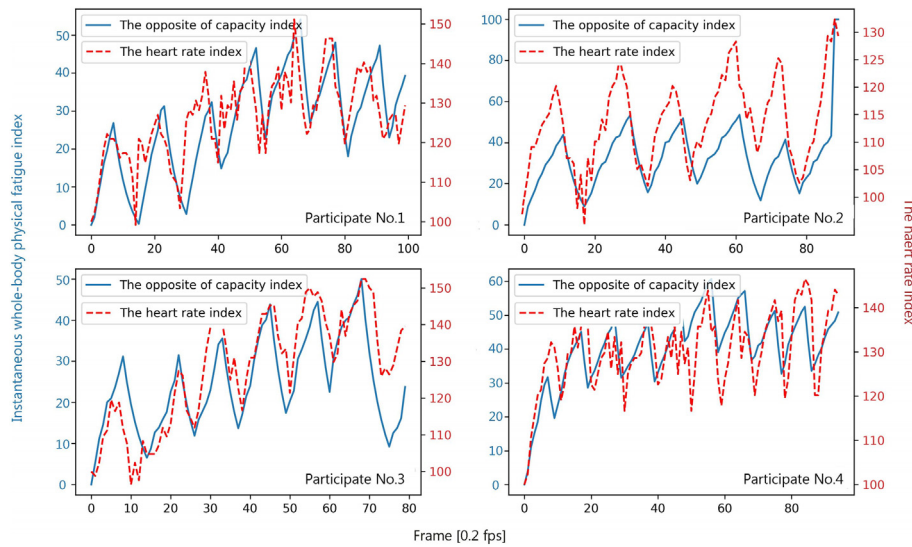


Fig. 8. The comparisons of the average instantaneous physical fatigue index and the normalized heart rate of the four participants.

5. Discussion

This study, for the first time, provides a non-invasive and automatic physical fatigue assessment method for construction site workers by using deep learning algorithms, biomechanical analysis and a physical fatigue model. As mentioned in the Results, the method can provide objective joint-level fatigue assessments during the whole working process. The following provides two cases of applying the proposed fatigue assessment method in different construction tasks.

5.1. Use of the proposed fatigue assessment method in three construction tasks

5.1.1. Case #1: the influence of construction site layout on fatigue level

Different construction site layout may result in different working postures and working duration, and further lead to different fatigue level and productivity. Take the scaffolding task for example. If the distance between the working area and the storage area is increased, the worker needs to carry the tubes for a longer period, but he also gets more time to have a rest when he goes back to the storage area without



Fig. 9. The process of the scaffolding task and the finished cube.



Fig. 10. The site layout of the scaffolding task.

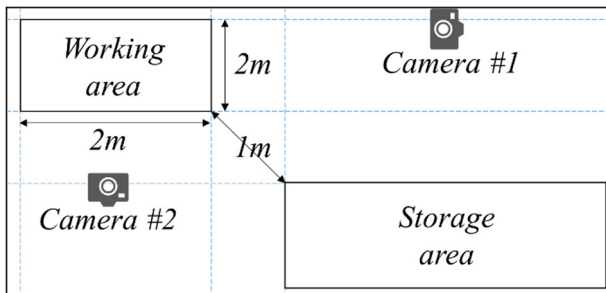


Fig. 11. The process of the masonry task and the finished concrete block wall.

Table 3

The anthropological parameters of the experiment subjects.

Subject	Height [m]	Weight [kg]	Age	Gender
The scaffolder	1.70	75	27	Male
The masonry	1.75	71	40	Male

Table 4

The information of the video records of the two experiment tasks.

Task	Duration [second]	Frame size	Data rate	Frame rate [fps]	No. of frames
Scaffolding	1433	640 × 480	4500 kbps	30	42,990
Masonry	378				11,330

any external load. As a result, the influence of construction site layout on the worker's physical fatigue level seems to be ambiguous. The proposed physical fatigue assessment method could help to provide an objective comparison between different site layouts, as shown in the following experiments.

In this experiment, the distance between the working area and the storage area was set as 3 m (half of the original distance) and 12 m (double of the original distance). Figs. 13a and 13b show the

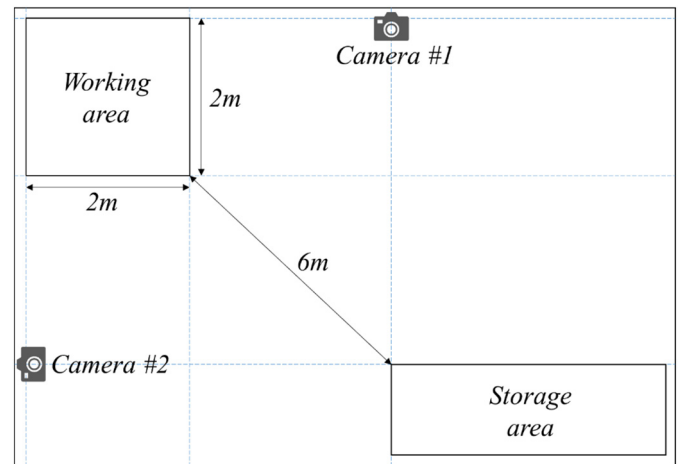


Fig. 12. The site layout of the masonry task.

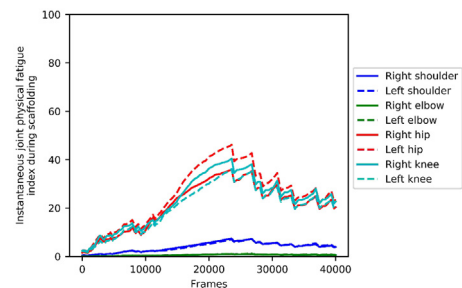


Fig. 13a. The instantaneous joint physical fatigue index of the key joints during the scaffolding task.

comparison results. The vertical axes represent the average joint capacity and fatigue index of the eight joints. The horizontal axes represent the frames. Figs. 13a and 13b reveal that the longer distance

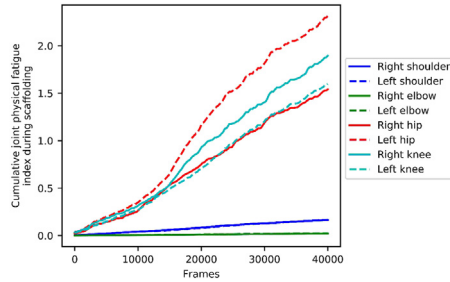


Fig. 13b. The cumulative joint physical fatigue index of the key joints during the scaffolding task.

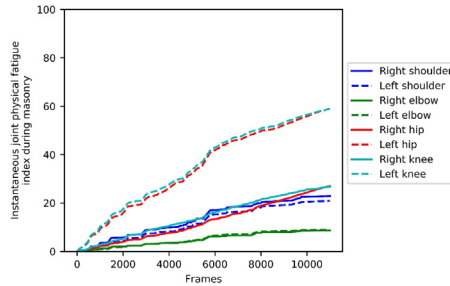


Fig. 13c. The instantaneous joint physical fatigue index of the key joints during the masonry task.

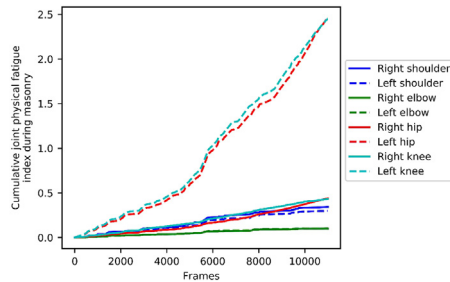


Fig. 13d. The cumulative joint physical fatigue index of the key joints during the masonry task.

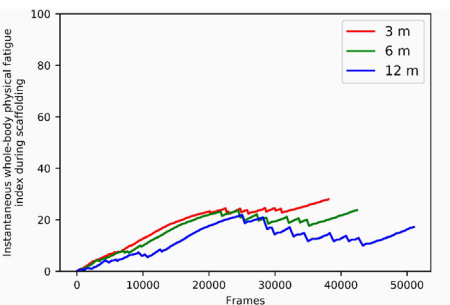


Fig. 14a. The comparison of the instantaneous whole-body physical fatigue indices during the scaffolding task with different distances between the working area and storage area (3 m/6 m/12 m).

between the worker area and the storage area, the lower the fatigue level, which mean that the influence of increased rest time demonstrated the influence of increased load-carrying time. Figs. 13c and 13d provide a comparison in the final fatigue level and the duration of the scaffolding task between the three site layout plans. With the increasement of the distance, the final fatigue index decreased, while the working duration increased. As a result, in the site layout design of this case, there existed a trade-off between increasing productivity and decreasing fatigue level.

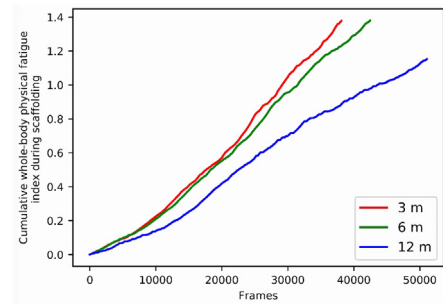


Fig. 14b. The comparison of the cumulative whole-body physical fatigue indices during the scaffolding task with different distances between the working area and storage area (3 m/6 m/12 m).

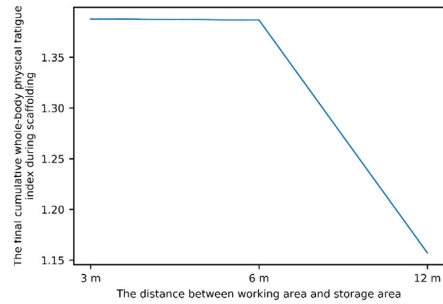


Fig. 14c. The comparison of the final cumulative whole-body physical fatigue indices after the scaffolding task with different distances between the working area and storage area (3 m/6 m/12 m).

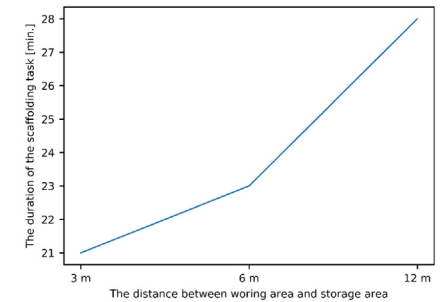


Fig. 14d. The comparison of the duration of the scaffolding task with different distances between the working area and storage area (3 m/6 m/12 m).

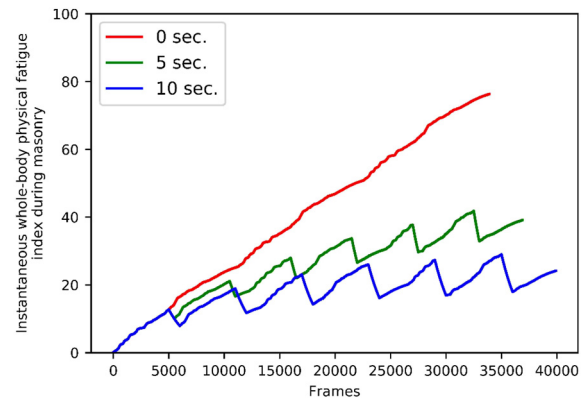


Fig. 15. The comparison of the instantaneous whole-body physical fatigue indices during the masonry task with different rest time (0/5/10 s).

5.1.2. Case #2: the influence of work-rest schedule on fatigue level

Having a rest is an effective way to mitigate physical fatigue. However, construction workers usually don't have enough time to have

a rest during working. This experiment aims to validate the positive effects of rest on fatigue mitigation with quantitative fatigue assessment results.

In the original masonry task, the worker worked continuously with no rest. In this experiment, the worker had a rest after finishing a layer. The rest time was set as 1 s, 3 s and 5 s. The fatigue assessment results were shown in Figs. 14a–14d. The vertical axis represents the fatigue level with the average capacity of the eight joints. Compared with continuous working, having a rest could significantly decrease the fatigue level: After continuous working, the worker's average joint capacity remained only 25%. However, with a rest of 1 s, 3 s and 5 s, the worker's average joint capacity remained approximately 35%, 60% and 75%. In this case, even a 5-second rest could help the worker to recover his physical capacity and decrease the fatigue level (Fig. 15).

6. Limitations and future works

Above experiments demonstrate the potential of the proposed method in the fatigue assessment for construction workers. For construction site workers, such a fatigue assessment method could help them to understand the fatigue level of each body segment during various construction tasks. For construction site managers, the fatigue assessment results could assist them to evaluate the safety and health risks in different construction tasks arrangements.

Despite numerous advantages, the current study has a few limitations. First, it is important to provide accurate 3D motion estimation for fatigue assessment. However, the 3D motion estimation method in this research cannot provide accurate 3D motion estimation when there are severe vision obstructions or under top-down perspectives. Accordingly, future studies should train a 3D motion estimation model with more on-site pictures, especially those with obstructions and top-down perspectives. Second, the current method needs to measure the mass of materials or tools before working, which may limit the method's applicability in real construction projects. Future studies should focus on the automatic mass measuring methods, such as first identifying the material or the tool and then estimating the mass based on the identification results. Additionally, this study identifies the worker's work/rest status manually. Accordingly, future studies should focus on identifying the work/rest state automatically based on joint torque or postures.

7. Conclusions

The current study adopted a 3-step method to assessment construction workers' physical fatigue non-intrusively and automatically. First, a 3D motion estimation method was used to gather the worker's 3D motion data with a monocular RGB camera. The study found that the 3D motion estimation method could successfully infer the worker's 3D joint locations with an error of about 4 cm. Second, kinetics and kinematics were used to calculate the joint torques. Last, a fatigue model was used to calculate the current joint capacity and provide a fatigue index based on the history of joint capacity. The proposed method has a great potential to enhance the construction worker's understanding of the physical fatigue in construction tasks, and to facilitate the managers to consider construction workers' fatigue when arranging construction tasks.

Acknowledgements

This work is supported by the General Research Fund project funded by the Research Grant Council of the Hong Kong University Grants Committee under Grant No. 152099/18E.

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